

LUND
UNIVERSITY

The sharp A_p bound for the Hardy–Littlewood maximal operator

SIMON GUSTAFSSON





Weights and weighted L^p spaces

Definition 1 (Weights)

A *weight* is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$ that is not almost everywhere zero.

Definition 2 (Weighted L^p spaces)

Let w be a weight and let $1 \leq p < \infty$. The weighted L^p space is

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(y)|^p \, w(y) \, dy < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(y)|^p \, w(y) \, dy\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

LUND UNIVERSITY



LUND
UNIVERSITY